SAN capacity review Q4

Where the growth went and what we buy next

SAN TEAM · NOVEMBER 2026

Agenda

1. Where we are
2. What grew
3. Risks
4. The ask

We have 14 months of space: reclaim now, expand in Q1

EXECUTIVE SUMMARY

Capacity is **78% used**. Reclaiming old copies buys 5 months; the Q1 expansion buys 2 years.

1. **PMAX-A drives the growth**
   40 TB a month since July, half of all new capacity.

2. **90 TB can be reclaimed**
   Unmapped LUNs and old database refresh copies.

3. **Order by January**
   10 weeks lead time; online before the 80% line in March.

The ask: Approve the reclaim plan and the Q1 expansion budget.

01



Where we are

Four arrays, two sites

WHERE WE ARE

Three arrays healthy, **ONTAP-2 replication failed**

PMAX-A

Warning

78%

Used; 80% line in March

PMAX-B

OK

55%

Used; SRDF in sync

ONTAP-1

OK

64%

SnapMirror lag 12 min

ONTAP-2

Failed

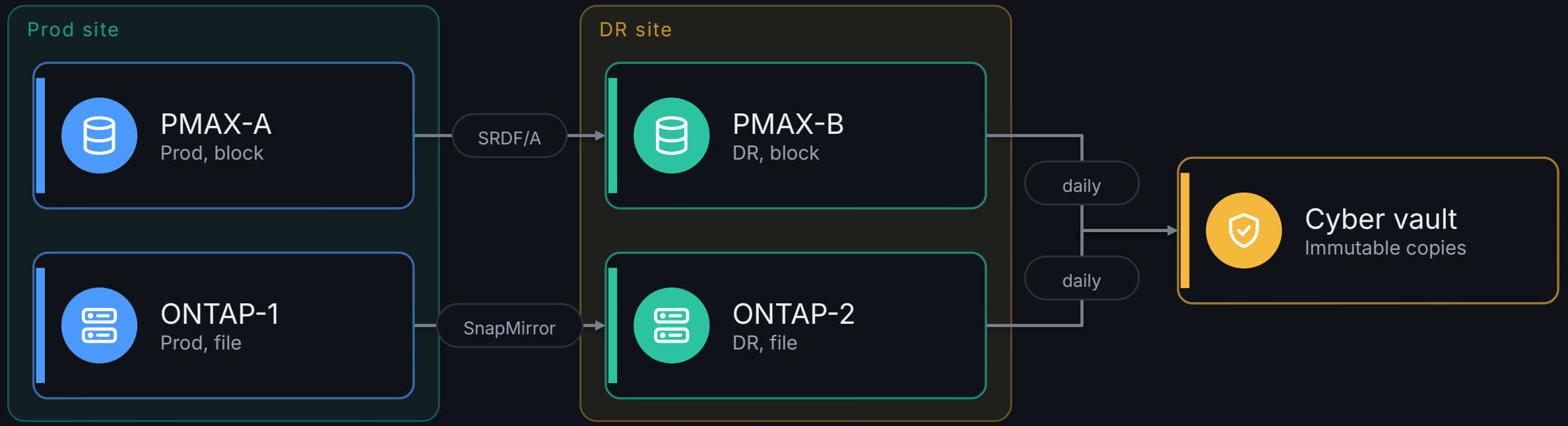
41%

SnapMirror broken since 2026-10-30

Source: Unisphere and ONTAP System Manager, 2026-11-01

WHERE WE ARE

Prod replicates to DR; DR copies go to the vault



One broken link (ONTAP-1 to ONTAP-2) leaves file data without a DR copy.

WHERE WE ARE

Usable capacity is **78% used** across the four arrays

1.9 PB

usable capacity

78%



used

▲ 6 pts in 90 days

4.1:1

data reduction

14

months to full at this rate

Source: Unisphere, 2026-11-01

WHERE WE ARE



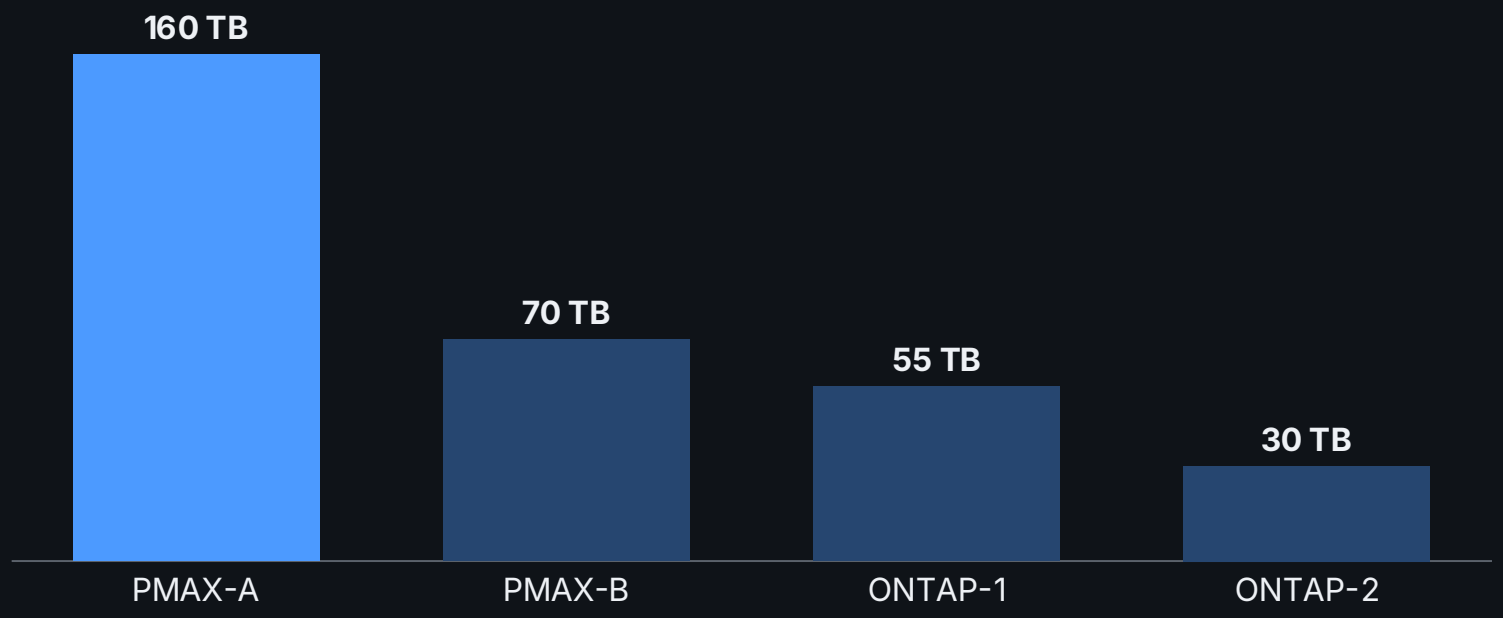
14 months

until the arrays are full at today's growth

Unless we reclaim or expand

WHERE WE ARE

Prod-A grew fastest: 40 TB a month since July



Source: Unisphere / Harvest

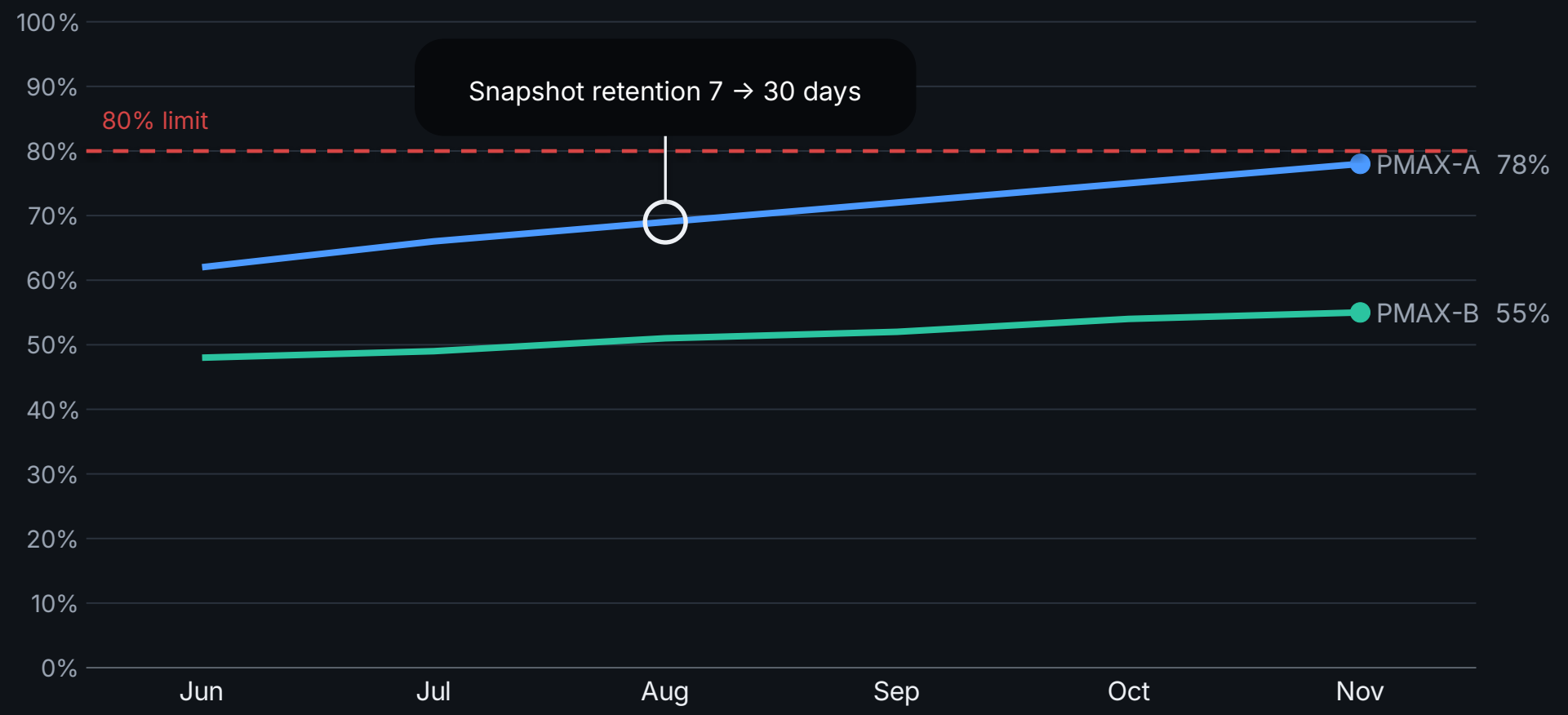
SAN team · capacity review

WHAT IT SHOWS

Prod-A takes **half** of all new capacity; the database refresh drives it.

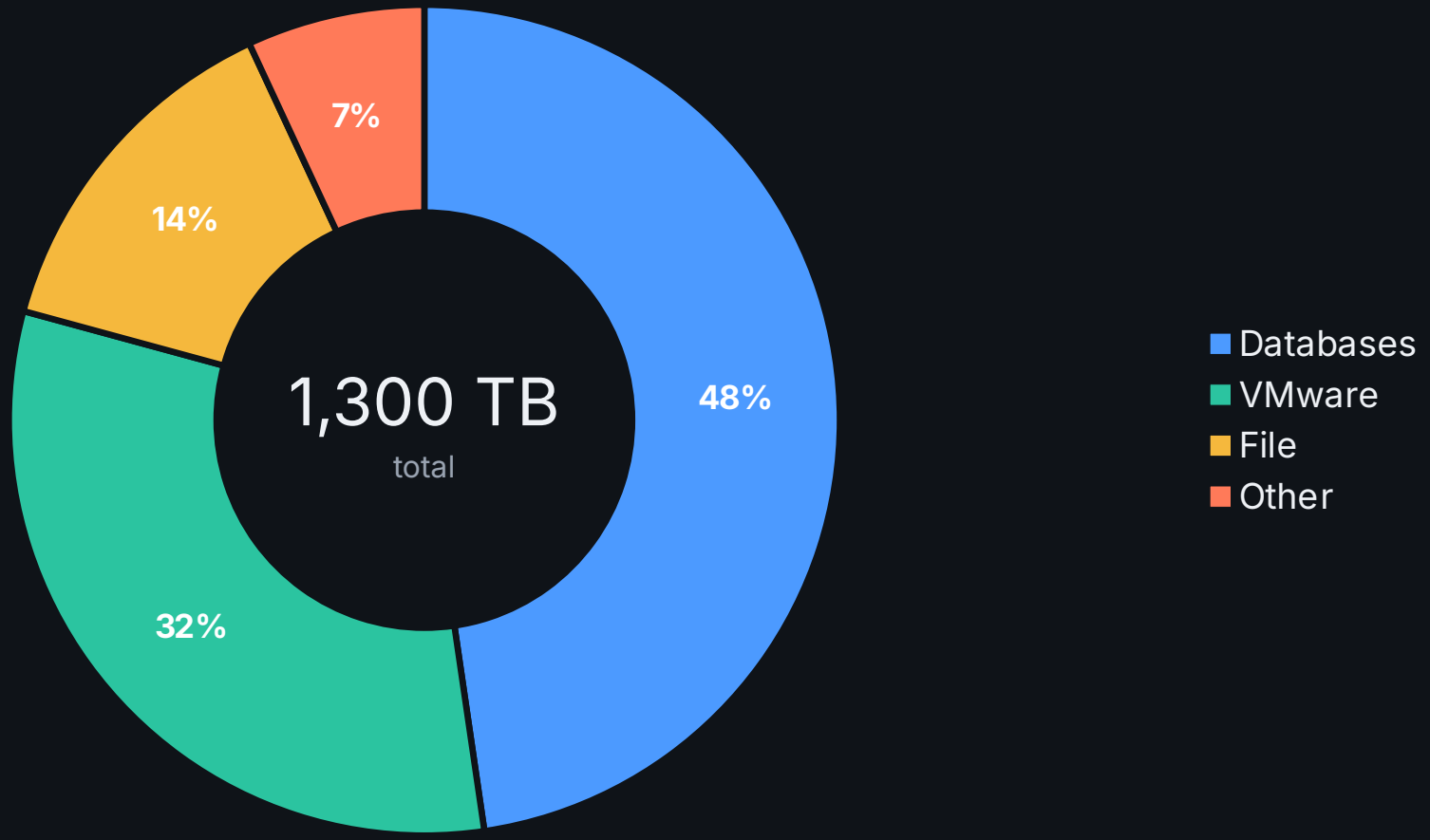
WHERE WE ARE

PMAX-A reaches the 80% line in March



WHERE WE ARE

Half of the space is databases



WHAT IT SHOWS

Databases grow 3% a month; everything else is flat.

WHERE WE ARE

Three things drive the growth

- **Database refresh** copies on Prod-A: 12 new 4 TB LUNs
- Snapshot retention moved from 7 to 30 days
  - policy change in August
- Two new VMware clusters on ONTAP-1

KEY POINT

Reclaiming old refresh copies frees about **90 TB** without buying anything.

WHERE WE ARE

Three ways to win back capacity



90 TB

Reclaim

Remove unmapped LUNs and old refresh copies.



60 TB

Rebalance

Move test workloads from PMAX-A to PMAX-B.



300 TB

Expand

Add one engine of capacity to PMAX-A in Q1.

WHERE WE ARE

What we check before a reclaim

No I/O for 90 days

Unisphere and host-side, both.

The owner agrees

In the change ticket, in writing.

A snapshot is kept

14 days, so an unmap can be undone.

The space is recorded

Freed TB per array in the capacity sheet.

WHERE WE ARE

Reclaim is quick and cheap; expansion lasts longest

Speed →

Quick, small

Rebalance test workloads: 60 TB in December.

Quick, large

Reclaim old copies: 90 TB in 3 weeks.

Slow, small

Tune data reduction: a few TB.

Slow, large

Expand PMAX-A: 300 TB by March.

Effect →

WHERE WE ARE

Reclaim + rebalance cut PMAX-A to **63%**

BEFORE

78%

PMAX-A used today

-15 pts

AFTER

63%

after reclaim and rebalance

That buys about 5 months before the expansion is needed.

WHERE WE ARE

Reclaim first, then expand

Reclaim + rebalance

- No purchase
- Done in 3 weeks
- Buys about 5 months

Expand now

- Purchase order and lead time
- Done in 10 weeks
- Buys 2+ years

Do both: reclaim now, order the expansion for Q1.

WHERE WE ARE

Arrays at a glance

Array	Site	Used %	Free TB	Replication
PMAX-A	Prod	78%	118	● In sync
PMAX-B	Prod	55%	290	● In sync
ONTAP-1	DR	64%	150	● Lagging
ONTAP-2	DR	41%	260	● Failed
Total		**62%**	**818**	

Replication: SRDF and SnapMirror state on 2026-11-01

WHERE WE ARE

Reclaim by December, new capacity by March



Reclaim

Unmap and remove, with owners' approval.

Rebalance

Move test SGs to PMAX-B.

Order

Expansion PO.

Install

New capacity online.

WHERE WE ARE

How a reclaim is approved



unused-luns report lists candidates.

Owner confirms within 5 days.

Change ticket, unmap, wait 14 days.

Remove and record the space freed.



The cheapest terabyte is the one you already own.

— Storage team motto

The ask

- Approve the reclaim plan
- Budget the Q1 expansion

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